9

Where AI Is Heading Today?

9.1 Expert System

An expert system is a computer program that solves complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise. That is, it is designed to provide decisions just like a human expert. Expert systems are the computer applications developed to solve complex problems.

Expert systems are designed for a specific domain, such as **medicine, science,** etc.

To perform its task, the expert system analyses user queries to extract knowledge from its knowledge base using the reasoning and inference rules.

The first expert system was developed in 1970. This system makes decisions for complex problems using **both facts and heuristics like a human expert.** The performance of an expert system depends on the knowledge stored in its knowledge base. More the amount of knowledge stored in the KB, better is the performance of the expert system. ***The suggestion of spelling errors while typing in the Google search box is a perfect example of expert systems used these days by all users.***

An expert system is not used to replace the human experts; instead, it is used to assist the human in making a complex decision.

The working of an expert system can be understood through the block diagram given in **Fig. 9.1**. The main components of an expert system include user interface, knowledge base and inference engine.

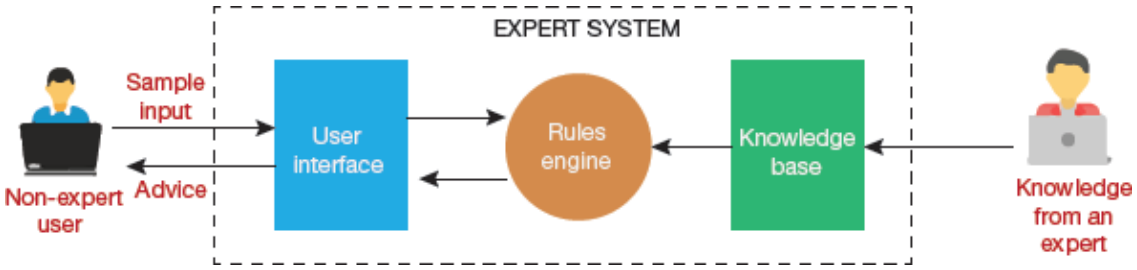


FIGURE 9.1 Working of an expert system

9.1.1 Popular Examples of the Expert System

1. **DENDRAL:** It is a chemical analysis expert system. DENDRAL was used in organic chemistry to detect unknown organic molecules with the help of their mass spectra and knowledge base of chemistry. The expert system studied a substance's spectrographic data to predict its molecular structure.
2. **MYCIN:** It is one of the earliest backward chaining expert systems that was designed to diagnose blood clotting diseases and find the bacteria causing infections like bacteraemia and meningitis. MYCIN could also recommend antibiotics and drugs based on the person's physical conditions.
3. **PXDES:** It is an expert system that determines the type and level of lung cancer by analysing picture(s) from the upper body.
4. **CaDeT:** CaDeT is an expert system that works like a diagnostic support system to detect cancer at early stages.
5. **R1/XCON:** The expert system could select specific software to generate a customized computer system based on user's requirements.
6. **DXplain:** It is a clinical support system that could suggest the presence of one or more diseases based on the findings of the doctor.

9.1.2 Characteristics of an Expert System

Expert systems are among the first truly successful forms of artificial intelligence (AI) software. Let us read about their distinguishing characteristics (refer **Fig. 9.2**).

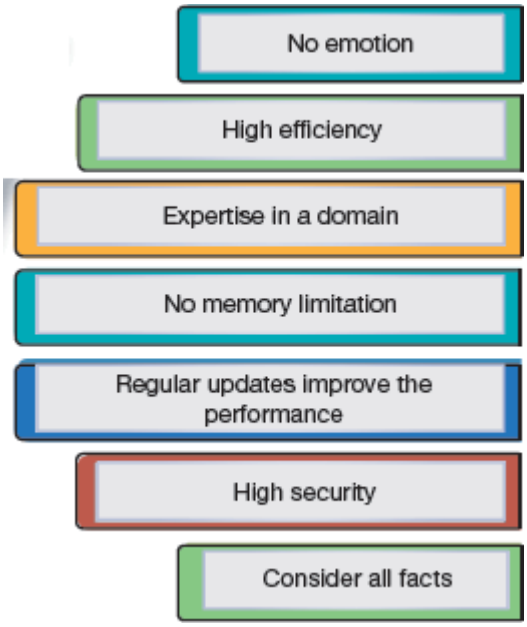


FIGURE 9.2

1. **High performance:** An expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy. The success of an expert system depends on its knowledge base. So, to get accurate knowledge continuously, the knowledge base is updated regularly.
2. **Understandable:** The results of an expert system are easily understood by users. The system takes input in human language and provides the output in the same language.
3. **Reliable:** An expert system generates efficient and accurate output. This makes it a reliable system.
4. **Highly responsive:** An expert system generates results for any complex query within a very short period of time, making it highly responsive. The total time by an expert system should be less than that taken by a human expert.
5. **No memory limitations:** An expert system can store and exploit huge amounts of data.
6. **Expertise in a domain:** There are many human experts in a particular domain and each of them specialize in a different skill and has different experiences. When we put the knowledge gained from these human experts into the expert system, the results are provided by utilizing all the facts and knowledge

7. **Not affected by emotions:** An expert system is not affected by human emotions such as fatigue, anger, depression, anxiety, etc. This helps them to give a consistent performance every time they are used.
8. **Not biased:** To respond to any query, the expert systems check and consider all the available facts.
9. **Reduce cost:** Human experts charge a heavy amount for examining, diagnosing and treating a patient. But with an expert system, multiple experts can be consulted at much cheaper rates.
10. **Non-perishable:** Human experts are perishable as they have a limited life span but an expert system is permanent. Once developed, it can be used without any limitations of expiry.
11. **Intelligent:** These systems use knowledge base and inference engine to solve complex problems. This is done by deducing new facts from existing facts that are represented as if-then rules.

9.1.3 Components of an Expert System

An expert system mainly consists of three components—**user interface, inference engine and knowledge base** (as shown in **Fig. 9.3**).

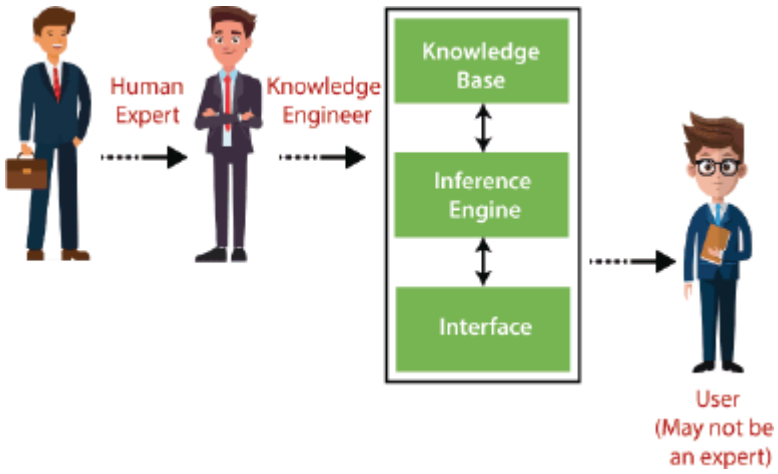


FIGURE 9.3 Components of an Expert System

User Interface

The user interacts with expert system through the user interface. It is through this interface that the user enters queries in a readable format. The interface then passes it to the inference engine. After getting response from the inference engine, it displays the output to the user. The

user interface is designed to be very simple and intuitive so that even a **non-expert user can communicate with the expert system to find solution to a complex problem.**

The user interface also explains how the expert system derived a particular conclusion. This explanation may be text displayed on screen, verbal narrations in natural language or a listing of rules on the screen. Through this, the user interface makes it easy to trace the credibility of the conclusions. An effective user interface must have the following features:

1. Help users to accomplish their goals in shortest possible way.
2. Be easy to customize as per user's requirements.
3. Make efficient use of user input.

Inference Engine (Rules of Engine)

The inference engine is the main processing unit and hence known as brain of the expert system. The engine derives conclusion or deduces new information by applying inference rules to the knowledge base. There are two types of inference engine:

Deterministic Inference Engine

This engine uses facts and rules to draw conclusions. These conclusions are thus assumed to be true.

Probabilistic Inference Engine

This inference engine contains uncertainty in conclusions as it is based on the probability.

Efficient procedures and rules used by the inference engine helps to formulate correct, flawless solutions. In case there is a conflict when multiple rules are applicable to a particular case, the inference engine resolves the problem by selecting the best rule that is applicable in the current context. Inference engines can also include explanation and debugging abilities. The explanation module, if incorporated, can be used by the expert system to give the user an explanation about how it reached a particular conclusion.

The inference engine generally uses two strategies for acquiring knowledge from the knowledge base and deriving solutions. These are as follows:

1. **Forward Chaining:** As discussed earlier, forward chaining starts from the known facts and rules (refer **Fig. 9.4**). After applying the inference rules, conclusions are drawn and then added to the known facts. Forward chaining strategy facilitates an expert system to answer the question, **'What can happen next?'** For example, forward chaining can be used to predict stock prices after changes in interest rates.

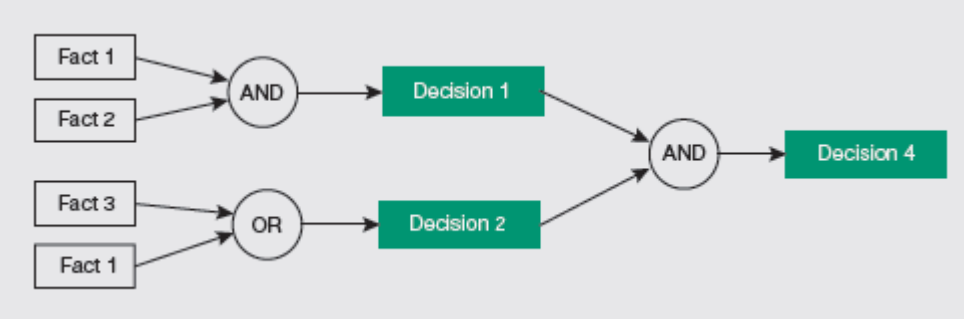


FIGURE 9.4 Forward Chaining

2. **Backward Chaining:** Backward reasoning, on the other hand, starts from the goal and works backward to prove the known facts (as shown in **Fig. 9.5**). This strategy enables an expert system to answer the question, **'Why this happened?'**
   If an event has already taken place, then the inference engine can be used to find out which conditions occurred in the past for this result. Thus, backward chaining strategy is used for discovering cause or reason. For example, diagnosis of blood cancer in humans.

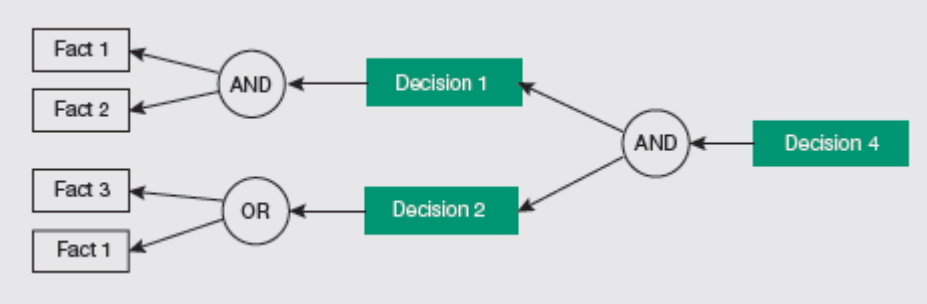


FIGURE 9.5 Backward Chaining

Knowledge Base

Before going to knowledge base, let us first understand what knowledge is. From junior classes, we are taught that data is a collection of facts. Data, when processed, gives us information. Data, information and past experience combined together is known as knowledge.

We have read earlier that the knowledge base stores knowledge acquired from different experts of a particular domain. More the knowledge in the knowledge base, more precise is the result from the expert system.

The knowledge base contains information and rules of a particular domain or subject. The two main components of a knowledge base are factual knowledge and heuristic knowledge.

Factual knowledge is based on facts and accepted by knowledge engineers in that domain.

Heuristic knowledge is based on practice, probability, evaluation and experiences.

Knowledge in a knowledge base is usually represented using the If-else rules and is acquired by extracting, organizing and structuring the domain knowledge under the guidance of human experts. This knowledge is domain-specific and of high-quality.

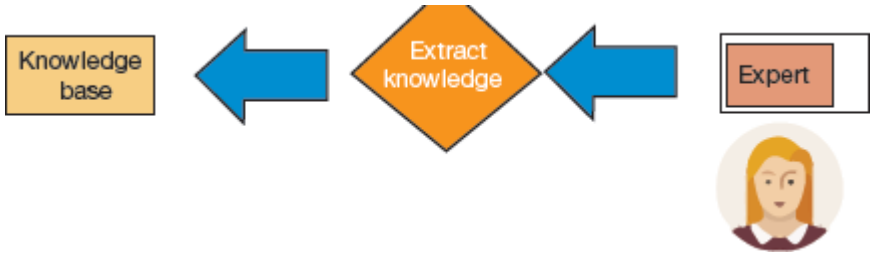


FIGURE 9.6 Creation of Knowledge Base

CASE STUDY: Working of MYCIN as an expert system.

Step 1: Knowledge base is created with expert knowledge about bacterial infection—their causes, symptoms, treatment, etc. (refer **Fig. 9.6**).

Step 2: The knowledge base is regularly updated. In order to test it, the doctor gives the expert system a new problem of identifying the presence of the bacteria by inputting the details of a patient, including the symptoms, current condition, and medical history.

Step 3: The system prompts the patient to provide general information like gender, age, etc.

Step 4: The system applies if-then rules using the inference engine and the facts stored within the knowledge base.

Step 5: The output is displayed to the patient through the user interface.

9.1.4 Participants in the Development of Expert System

In **Fig. 9,7**, we see that there are three primary participants in an expert system:

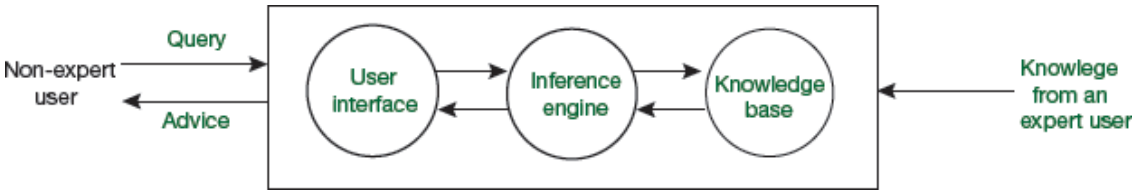


FIGURE 9.7 Participants in an expert system

Expert

The performance of any expert system depends on the knowledge fed in its knowledge base. This knowledge is provided by human experts who are specialized in that specific domain.

Knowledge Engineer

The term ‘knowledge engineering’ refers to the process of building an expert system and its practitioners are called knowledge engineers. The main task of a knowledge engineer is to ensure that the computer possesses all the knowledge required to solve a problem. For this, he/she gathers the knowledge from the domain experts and then represent that knowledge in the format accepted by the knowledge base.

End User

The end user who uses the expert system may or may not be well versed in that domain.

He/she just uses the expert system to find solution or advice for his queries, which are usually complex.

9.1.5 Capabilities of the Expert System

An expert system can do the following things:

1. **Advise:** The system can give its advice to users for any query—simple or complex— that fits its domain.
2. **Provide decision-making capabilities:** An expert system can be designed to make complex decisions in any field, be it finance, medical science or any other.
3. **Demonstrate a device:** It is can be used to demonstrate features, specifications, usage of any new product.
4. **Problem-solving:** It can be used to solve complex problems.
5. **Explaining a problem:** An expert system can be used to provide a detailed description of an input problem.
6. **Interpreting the input:** It can interpret the input given by the user.
7. **Predicting results:** An expert system can predict results using past data.
8. ***Instructing and assisting human in decision-making***
9. ***Diagnosing:*** Expert systems, especially in the field of medical science, are used to diagnose any disease.
10. ***Justifying the conclusion:*** Expert systems not only find solutions to complex real-world problems by drawing conclusions from known facts but also justify those conclusions.
11. ***Suggesting:*** The best part is that an expert system also suggests alternative options to a problem.

However, expert systems cannot do the following:

1. Substitute human decision-makers
2. Possess human capabilities

3. Produce accurate output for inadequate knowledge base
4. Refining knowledge base
5. Use emotions to make decisions

9.1.6 Advantages of Expert Systems

1. Results obtained by an expert system are highly reproducible.
2. They can be used in risky places where it is not safe for humans to work.
3. They give accurate results, especially when the knowledge base is updated regularly.
4. They give consistent performance at a fast speed as these systems are not affected by emotions, tension or fatigue.
5. They are easily available due to mass production of software.
6. They are affordable as the production cost is reasonable.
7. They work at great speed and reduce human effort.
8. They are capable of explaining how the solution was obtained.
9. They improve the quality of decisions.
10. They gather scarce expertise knowledge in good amount and use it efficiently.

9.1.7 Limitations of Expert Systems

1. Results obtained from an expert system may get wrong if the knowledge base contains wrong/obsolete information.
2. Each problem is different; therefore, the solution from a human expert may be strikingly different from that of an expert system.
3. They lack creativity for different scenarios, especially in extraordinary situations.
4. They have high development and maintenance costs.
5. Knowledge acquisition is a challenging task.
6. Success depends on knowledge acquired from human experts.
7. These systems cannot learn from themselves and hence require manual updates.
8. They cannot produce correct results from less amount of knowledge.
9. They need excessive training.
10. They do not have any emotions.
11. They do not possess any common sense.

12. They are developed only for a particular domain.
13. They are not capable of making decisions in extraordinary situations.
14. They work on Garbage-in Garbage-out (GIGO) principle. This means that if there is an error in the knowledge base, the expert system will give incorrect results.
15. They have high maintenance cost.

9.1.8 Applications of Expert Systems

Expert systems are one of the prominent research domains of AI. Some common applications of expert system are as follows:

1. **Designing and manufacturing** devices such as VLSI systems, camera lenses and automobiles.
2. **P**ublishing relevant knowledge content to users. An advisor and a tax advisor are two popularly used expert systems in this domain.
3. **F**inance industries use expert systems to detect any type of possible fraud, suspicious activity, stock market trading and advise bankers on whether they should approve the loans or not.
4. **Expert systems are extensively used for medical diagnosis and treatment. In fact, it** was the first area where these systems were used.
5. **Expert systems are used for planning and scheduling** tasks for achieving the project's goal. Companies also use expert systems for scheduling airlines, cargo customer order, computer resources and various manufacturing task.
6. Tracking the progress of a software development project.
7. **Monitoring applications c**ompare data continuously with observed system or with prescribed behaviour to detect leakage in long petroleum pipeline using expert systems.
8. Process control systems use expert systems to control a physical process based by continuously monitoring it.
9. The automobile and electronics industries use expert systems to identify faults in vehicles and computers manufactured.
10. Diagnosing complex electronic, electromechanical and diesel-electric locomotive system.
11. Forecasting crop damage or in general predicting results of any experiment/activity.

12. Identifying the structure of a chemical compound.
13. Assessing geologic structure from dip meter logs and space structure through satellite and robot.
14. Evaluating civil cases, product liability, employee performance, etc. using logs.
15. Detecting virus attack in a computer system.
16. Repair and maintenance projects.
17. Optimizing warehouses.
18. Configurating manufactured objects.

9.1.9 Expert System Technology

Expert systems technologies include the following.

1. **Expert System Development Environment:** The development environment of an expert system includes hardware and tools like
    1. Workstations, minicomputers, mainframes.
    2. High level symbolic programming languages (like **LIS**t **P**rogramming (LISP) and **PRO**grammation en **LOG**ique (PROLOG)).
    3. Large databases.
2. **Tools:** The cost of developing an expert system gets reduced to a large extent with the help of tools that help developers by providing,
    1. Powerful editors
    2. Debugging capabilities with multi-windows
    3. Rapid prototyping
    4. Having in-built definitions of model, knowledge representation and inference design.
3. **Shells**: A shell is an expert system without knowledge base. It facilitates developers by providing modules for knowledge acquisition, inference engine, user interface and explanation facility. Two popularly used shells are JESS and Vidwan. While Java Expert System Shell (JESS) provides Java API for creating an expert system, *Vidwan*, enables knowledge encoding in the form of IF-THEN rules.

9.1.10 Development of Expert Systems

The process of developing an expert system is iterative. The steps to develop such a system are described below.

Identify Problem Domain

The first step to develop any system is to identify the problem that it intends to solve. Ensure that this problem is solvable by an expert system. If the problem can be solved, identify the human experts in the task domain and analyse the cost-effectiveness of the system. Knowledge engineer(s) and domain experts work in coherence to define the problem and list the characteristics of the system.

Design the System

To design the expert system, first identify the technology (hardware, tools and shells) that will be used. Understand how well the system will be integrated with the other systems and databases. Prepare a framework for representing concepts in domain knowledge.

Develop the Prototype

In this step, the knowledge engineer acquires domain knowledge from the human experts and represent it in the form of If-THEN-ELSE rules. The knowledge expert also determines how heuristic knowledge could be integrated in the reasoning process and what type of explanation would be useful.

Test and Refine the Prototype

The knowledge engineer tests the prototype to identify any deficiencies in performance or results by using sample cases. Testing is also done along with end-users to unleash additional errors or discrepancies. The results of testing are then used to refine the system.

Develop and Complete the System

Once the expert system is developed and tested, it is then tested along with other elements of its environment, including end users, databases and other information systems to ensure that it interacts in the prescribed manner and smoothly integrates with them to render the desired functionality. At this step, the expert system is clearly documented and users are trained for its usage.

An expert system is also used for information management and help desks management.

Maintain the System

Maintenance is an ongoing process. In this phase, the knowledge base is regularly reviewed and updated. New requirements from the users or blending with new interfaces of other information systems as those systems evolve are steps that need to be done time to time.

TABLE 9.1 Traditional System vs Expert System

Conventional System	Expert System
Knowledge and processing are combined in one unit.	Knowledge database and the processing mechanism are two separate components.
The program does not make errors (unless there is an error in input data or programming logic).	The expert system may give less accurate results.
The system is operational only when it is fully developed.	The expert system is optimized on an ongoing basis and can be used even with a small number of rules.
Program is executed step-by-step according to a fixed algorithm.	Execution is done logically and heuristically.
The problem expertise is encoded in both program as well as data structures	Problem-related expertise is encoded only in data structures.
It needs full information.	It can be functional even with insufficient information.
It uses data more efficiently.	It utilizes knowledge efficiently.
It gives solution without explanation.	It gives solution and explains how it was obtained.
It does not represent knowledge symbolically for computations.	It symbolically represents knowledge for computations.

TABLE 9.2 Human expert vs Expert System

Human Expert	Artificial Expertise
Perishable	Permanent
Difficult to transfer	Transferable
Difficult to document	Easy to document
Unpredictable	Consistent
Expensive	Cost-effective system

9.2 Internet of Things

Although the concept was not named until 1999, the Internet of Things (IoT) has been in development for decades. IoT means a system of interrelated computing devices, machines, objects, animals or people that have unique identifiers for identification and also the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction (as shown in **Fig. 9.8**).



Credit: 363868 / Shutterstock

FIGURE 9.8 Interconnected Things

If the thing, in the IoT, is a person, then he can have heart monitor implant; if it is an animal, it can have a biochip transponder, if it is an automobile, it can have sensors to generate alert alarms (e.g., when the pressure in a tyre is too low) and if it is any other natural or man-made object, then it is assigned an IP address and provided with the ability to transfer data over a network.

The huge availability of address space in IPv6 has led to the development of the IoT. With more IP addresses available, more devices can be connected to exchange information with each other.

Today, IoT is being widely used for precision agriculture, building management, healthcare, energy and transportation. The first Internet appliance was a Coke machine at Carnegie Melon University in the early 1980s when the programmers succeeded in connecting the machine over the Internet to check the status of the machine and determine the availability of a cold drink. This check helped them to save their time as they did not have to go to the machine to buy Coke when it was not available in the machine.

Later, other devices like cell phones, coffee makers, washing machines, headphones, lamps, wearable devices and almost anything you can think of could be connected. It is interesting to know that in 2022, the market for the IoT is expected to grow 18% to 14.4 billion active connections and by 2025, there will be approximately 27 billion connected IoT devices. So, we can conclude that, IoT is a giant network of connected ‘things’ (including people) which will have relationships between people-people, people-things and things-things.

In simple terminology, IoT is the concept of connecting any devices over the Internet.

9.2.1 Examples of Applications of IoT

Whenever we read about a new technology, the first question that comes to our mind is how is this technology going to help us? While with IoT, anything that can be connected, will be connected, but the question is why should we want these devices to be connected? There are many examples to justify the answers to these questions.

Case 1: Imagine you are going for meeting and your car tells you the best route to take. If there is a traffic jam, then your car sends a text message to the other party, notifying them that you will be late.

Case 2: Imagine that the moment your alarm rings to wake you, a message is sent to the coffee machine to start brewing coffee for you.

Case 3: The smart watch that your wear in office tells you when and where you were most active and productive.

Case 4: You have a self-driving car with complex sensors to detect objects in their path.

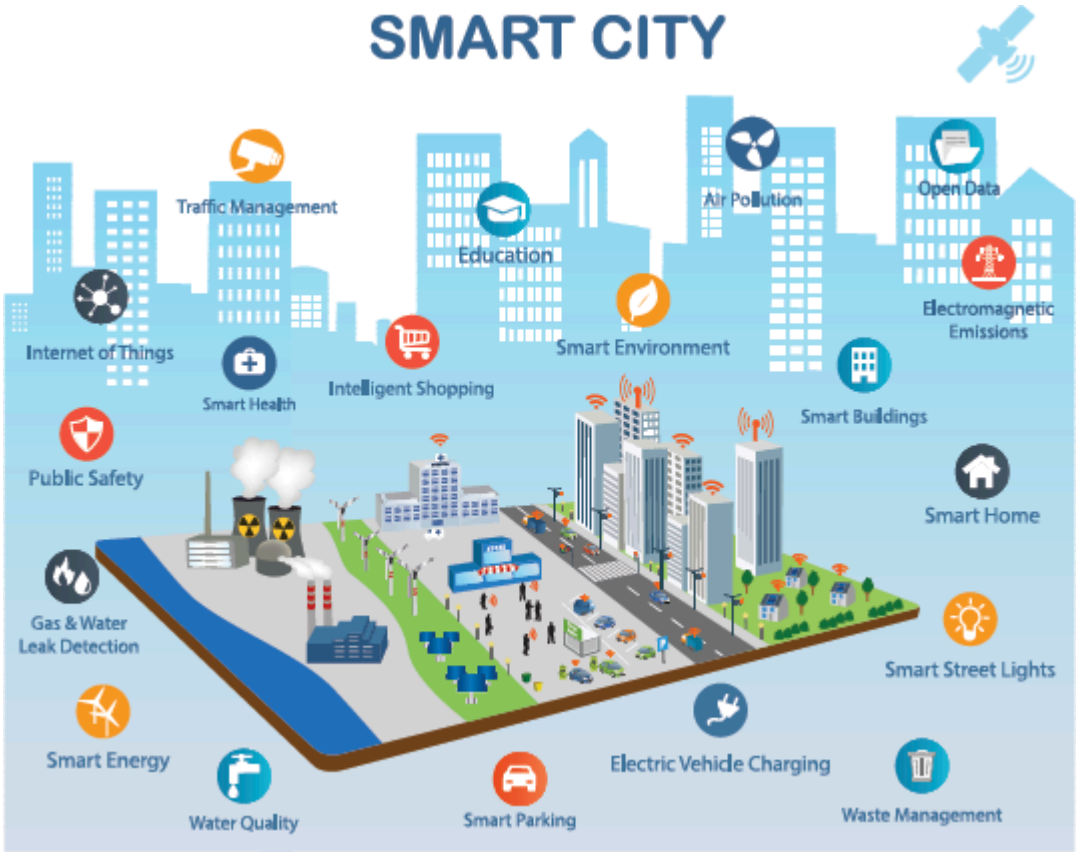
In 2019, IoT devices generated an estimated 18.3 zettabytes of data, which is expected to grow to 73.1 ZB by 2025.

Case 5: A smart football that can track how far and fast it is thrown and record those statistics via an app for future training purposes.

Case 6: You have a smart refrigerator that can text message you that milk packets are over in the fridge and you need to buy them before you come home. Or, your fridge checking the expiry date of an ice-cream and notifying you that you should not eat it now.

Case 7: Imagine that you have an AC installed in your home which is connected with your smartphone. On a very hot day, you can instruct your AC to start and set the room temperature to 18 degrees before you finally reach the house.

In an article, Ashton wrote that if computers knew everything about things (using data gathered by them) without any help from us, then we would be able to track and count everything. It would also reduce waste, loss and cost. We would know exactly when things need to be replaced, repaired or recalled, and whether they were fresh or past their best.



Credit: monicaodo. Shutterstock

FIGURE 9.9 A Smart City

In this way, IoT can help organizations save a lot of money through improved process efficiency, asset utilization and enhanced productivity. With improved tracking of objects using sensors and connectivity, companies can better analyse them and make smart decisions. For example, if you own a car manufacturing company, then you can know which accessories are the most popular by using sensors to detect which areas in the showroom are the most popular, and where customers linger longest. You can even use the available sales data to identify which components are selling fastest and then automatically align sales data with supply, so that popular items never go out of stock.

The information collected by IoT devices can be used to detect patterns, make recommendations, and detect possible problems before they occur.

We have heard that the Indian government is working hard to develop smart cities. Do you know that smart cities use IoT for efficient utilization

of resources? **Figure 9.9** gives an overview of such a city. IoT has the potential to transform entire cities by solving real problems citizens which are faced by the people every day. With the proper connections and data, the IoT can solve traffic congestion issues and reduce noise, crime and pollution.

9.2.2 IoT Products

Big companies like Honeywell, Hitachi, GE, Cisco, AT&T, Apple, Google, IBM, Microsoft, Skyworks, Iridium Communications, Red Hat, Zebra Technologies, InterDigital are already playing in the market to realize the benefits of IoT. Given below are some examples.

Amazon Echo for Smart Home works through its voice assistant, Alexa. Users can talk to Alexa and give order to perform a variety of functions. For example, users can tell Alexa to play music, provide a weather report, get sports scores, order an Uber and do much more.

Fitbit One – Wearables tracks your steps, floors climbed, calories burned, and quality of sleep. The device wirelessly connects with computers and smartphones to transmit your fitness data in understandable charts to monitor your progress.

Barcelona - Smart Cities: The Barcelona city in Spain is one of the foremost smart cities in the world. It has implemented several IoT initiatives that have helped to enhance smart parking and the environment.

Amazon Web Services, Microsoft Azure, IBM's Watson, Cisco IoT Cloud Connect, Salesforce IoT Cloud, Oracle Integrated Cloud and GE Predix are popular IoT platforms.

AT&T - Connected Car: AT&T added 1.3 million cars to its network in the second quarter of 2016. With this, the count of total number of connected cars rose to 9.5 million.

Thus, we see that IoT allows for virtually endless opportunities and challenges thereby making it a hot topic for research. According to a report, nearly $6 trillion will be spent on IoT solutions over the next five years.

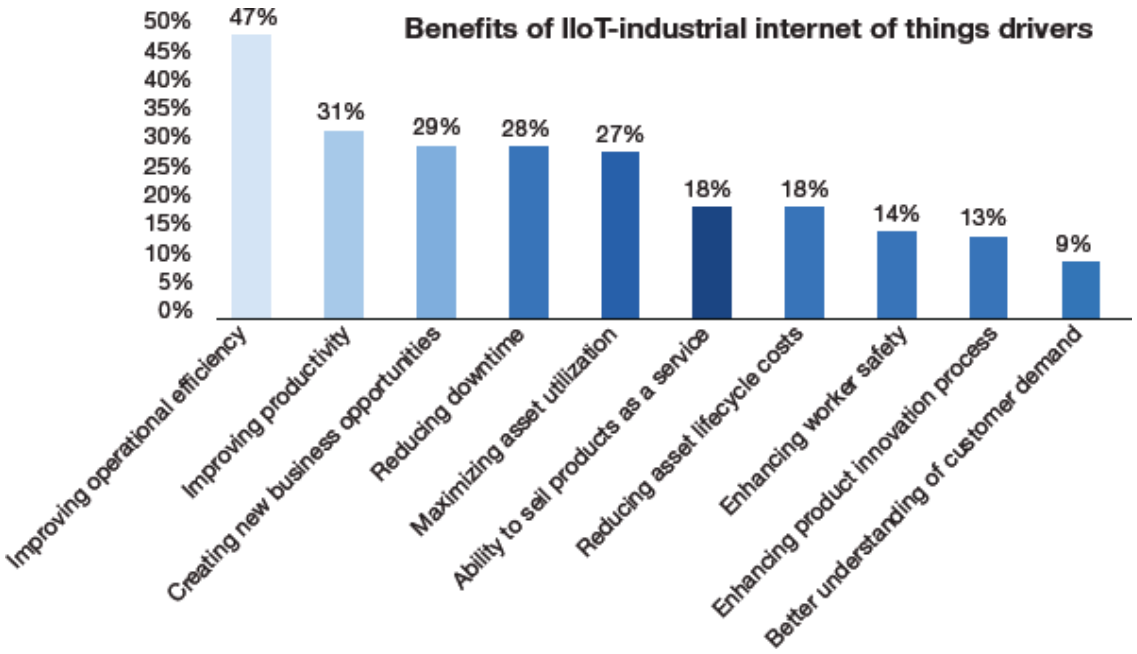


FIGURE 9.10 Benefits of IoT

9.2.3 Challenges

Security is the biggest issue that we often come across while talking about IoT. With billions of devices connected together, concerns of privacy and data sharing always haunt our minds.

In fact, the protection of sensitive data was ranked as the top concern among enterprises according to the 2016 Vormetric Data Threat Report. Hackers try to penetrate connected cars, critical infrastructure and even people's homes. Therefore, the main focus of companies is to ensure security of all the data generated by these devices. Most IoT devices do not encrypt communications if the data is transferred over a local Wi-Fi network. If the Wi-Fi network is unsecured, then it opens the gates for security threats wide open.

IoT devices used on patients as wearable devices, if left unattended (with technical errors), can be life-threatening for patient.

Another issue with IoT is that massive amounts of data is being generated by these devices. So, companies need to figure out how they would store, monitor, analyse and deduce results from this vast amount of data that is continuously being generated.

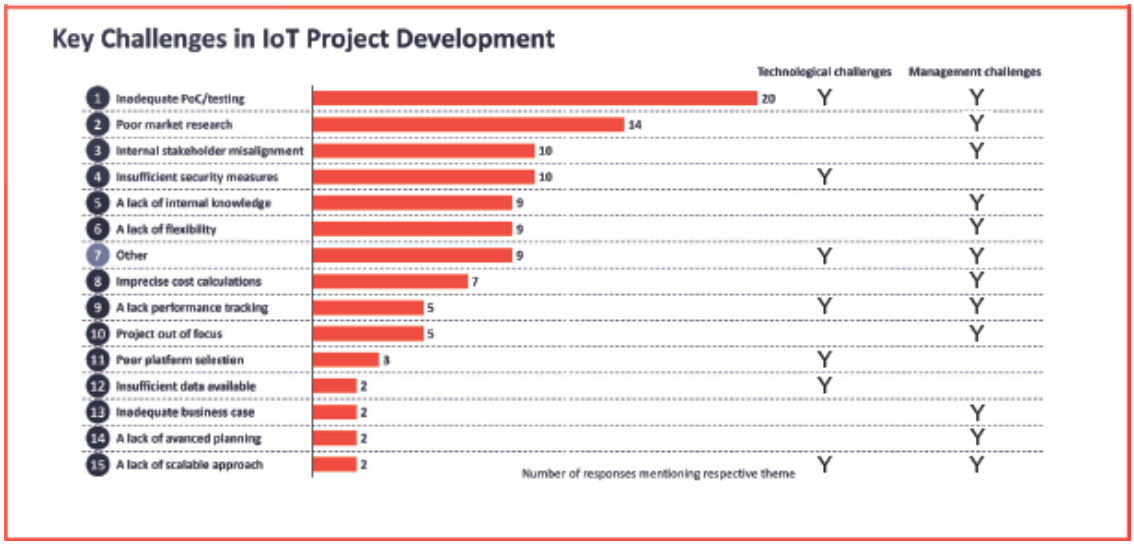


FIGURE 9.11 Key challenges in IOT project development

The graph in **Fig. 9.11** shows top challenges in the world of IoT. Besides security, other key concerns are as follows.

1. **Scalability: When b**illions of Internet-enabled devices are connected, large volumes of data are generated that need to be processed. To process this huge data, devices need big data analytics and cloud storage for interpretation of useful data. Moreover, these systems need to be scalable to accommodate ever expanding data.
2. **Interoperability:** Technological standards in most areas are still fragmented and need to be converged to establish a common framework and standard for the IoT devices.
3. **Lack of standardization:** Since standardization process is still lacking, interoperability of IoT with legacy devices is critical which, in turn, prevent us to move towards the vision of truly connected interoperable smart objects.
4. **Lack of government support:** Government and regulatory bodies like the FDA should set up regulations by forming a standard committee for safety and security of devices and people.

5. **IoT Malware and ransomware:** Ransomware can lock out users from various devices and platforms and still use a user's valuable data and information. For example, a hacker can hijack a computer camera and take pictures. He may also lock a device and then demand ransom to unlock it and return the data.
6. **Connectivity:** Poor Internet connectivity is a challenge where IoT sensors are required to monitor process data and supply information.
7. **Lack of effective and informed government regulations:** IoT is a fast-developing area of technology and many legislators are either not aware or do not fully understand the workings of IoT, so they are reluctant to create or enforce regulations.
8. **No privacy policies:** There are no clear regulations for IoT devices on which information is private or public. For example, an IoT toy could collect information from a child without taking consent from the parents. The toy can then share that data with the manufacturer who could then share it with others.

   IoT devices should be updated regularly. Updates should be small. Also, IoT devices should be backward compatible with older devices.
9. **Bandwidth strain:** As the number of IoT devices connected in a geographical area grows, more text/audio/video data will flow through those devices, thereby requiring higher bandwidth and placing strain on the server farms that process all that information.
10. **IoT professional skills gap:** IoT suffers from a professional skills gap. The main reason behind this is that IoT is growing rapidly and many people are yet to develop a skillset working with this technology. As of now, a few IoT experts are working in the industry, training new people even more difficult. In a survey, it was found that 38% of respondents reported that they lack in-house expertise in IoT network management, while 27% were facing a dearth of in-house experience to deploy IoT.

CASE STUDY EXAMPLE

Several high-profile IoT security breaches have been reported till date. One example is the 2015 Jeep Cherokee hack, when two hackers remotely took control of a car using vulnerabilities in the entertainment system to

access its dashboard functions. The hacker initiated a series of unexpected disturbances and finally disabled the brakes, causing the driver to swerve into a ditch.

Wireless IoT SIM cards should automatically update the system, removing the possibility of users not updating and leaving holes in system security.

CASE STUDY EXAMPLE

Inter-operability is a serious issue. Once it was seen that there was a problem when an insulin pump communicated only with a particular operating system. It makes the functionality unavailable for patients who may have a computer with a different operating system installed in it.

There are issues around the device's wireless standard. For example, if it uses Bluetooth, then the device could inadvertently connect to the wrong nearby device.

CASE STUDY EXAMPLE

A manufacturing plant where a large number of precise, high-resolution pictures are sent to 3-D printers could slow down the servers or even cause them to crash if the demands are too great.

Similarly, in a hospital, connected video cameras transfer of thousands of high-resolution radiology images and hundreds of telehealth video calls going on in a single day could significantly overload a server farm.

However, despite all odds, IoT is being used by several industries including manufacturing, defence, transportation, banks, retail, oil & gas mining, health care, connected home, banks to name a few.

9.2.4 Sensors

A sensor is a device that detects and responds to some type of input that it receives from the physical environment. This input could be heat, light, motion, pressure, moisture or any other environmental phenomena. As

the output, sensors usually generate a signal that is converted to human-readable form and then displayed at the sensor location or transmitted electronically over a network for reading or further processing.

For example, an oxygen sensor in a car's emission control system detects the gasoline/oxygen ratio. If the mixture is not optimal, the balance is readjusted.

Another example is motion sensors in home security lights, automatic doors and bathroom fixtures that send out microwaves, ultrasonic waves or light waves and detect when the flow of energy is interrupted by something entering its path.

Vision and imaging sensors detect the presence of objects or colours within their fields of view. They display a visual image of whatever it detects.

Temperature sensors detect thermal parameters and provide signals to the inputs of control and display devices. They are used to measure the thermal characteristics of gases, liquids and solids in many industrial processes.

Radiation sensors sense the presence of alpha, beta or gamma particles and provide signals to counters and display devices. They are usually used for surveys and sample counting.

Proximity sensors detect the presence of nearby objects through non-contacting means within a range of up to several millimetres. They are used in manufacturing operations to detect the presence of parts and machine components.

Pressure sensors detect forces per unit area in gases or liquids and provide signals to the inputs of control and display devices.

Position sensors sense the positions of valves, doors, throttles, etc. and supply signals to the inputs of control or display devices.

Photoelectric sensors sense objects passing within their field of detection. They can even detect colour, cleanliness and location, if required. Photo sensors are commonly used in manufacturing and material handling automation. For example, they are used for counting, robotic picking and automatic doors and gates.

Particle sensors sense dust and other airborne particulates and supply signals to the inputs of control or display devices. Particle sensors are common in bin and baghouse monitoring.

Motion sensors sense the movement or stoppage of parts, people, etc. and supply signals to the inputs of control or display devices. They are usually used for detecting the stalling of conveyors or the seizing of bearings.

Metal detectors sense the presence of metal in a variety of situations ranging from packages to people.

Level sensors are used to determine the height of gases, liquids or solids in tanks or bins. The information is then passed as signals to the inputs of control or display devices.

Leak sensors are used to identify or monitor the unwanted discharge of liquids or gases. Some leak detectors are used to measure the effectiveness of the seals in vacuum packages.

Humidity sensors measure the amount of water in the air and convert these measurements into signals that can be used as inputs to control or display devices.

Gas and chemical sensors sense the presence and properties of various gases or chemicals and relay signals to the inputs of controllers or visual displays.

Force sensors measure various parameters related to forces such as weight, torque, load, etc. and provide signals to the inputs of control or display devices.

Flow sensors sense the movement of gases, liquids or solids and provide signals to the inputs of control or display devices. They are used extensively in the processing industries.

Flaw sensors detect inconsistencies on surfaces or in underlying materials such as welds. They are used in a variety of manufacturing processes.

Flame detectors sense the presence and quality of fire and provide signals to the inputs of control devices. They are used in many combustion control applications like burners.

Electrical sensors sense current, voltage, etc. and provide signals to the inputs of control devices or visual displays.

Contact sensors detect physical touch or contact between the sensor and the object being observed or monitored. They are often used in alarm systems to monitor doors, windows and other access points. For example, when a door or window is opened/closed, a magnetic switch provides an indication to the alarm control unit so that the status of that entry point is known. They are also used as proximity sensors in robotics applications and automated machinery.

Non-contact sensors do not require a physical touch between the sensor and the object being monitored in order to function. Radar guns used by law enforcement to monitor the speed of vehicles is an example of a non-contact sensor.

Speed sensors detect the speed of an object or a vehicle.

Ultrasonic sensors detect noise and measure the distance between two objects. For this, high-frequency sound waves generated by active ultrasonic sensors are received back by the ultrasonic sensor for evaluating the echo. Time delay between transmitting and receiving the echo is used to calculate the distance to an object. Passive ultrasonic sensors are used for detecting ultrasonic noise present under specific conditions.

9.3 Artificial Intelligence of Things (AIoT)

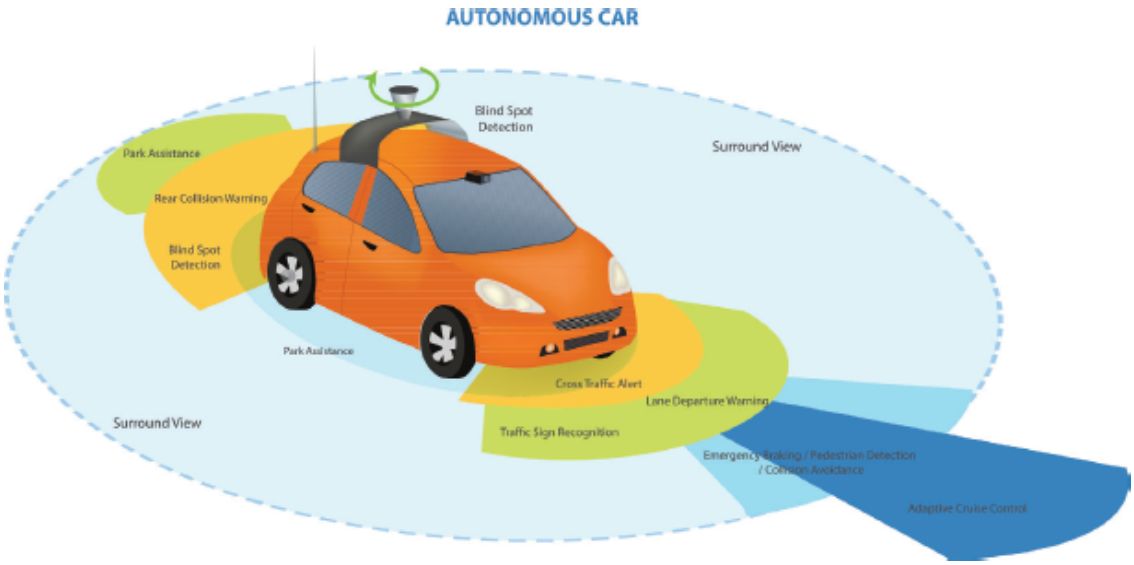
AIoT, or the combination of artificial intelligence (AI) technologies and the Internet of Things (IoT) infrastructure aims to make IoT operations more efficient, improve interactions between humans and machines and facilitates better data management and analytics.

We now clearly know that while AI simulates human intelligence processes by machines, IoT, on the other hand, is a system of interrelated computing devices with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. A *thing* or a computing device in IoT can be a person's heart monitor implant, a self-driving car with built-in sensors, a door mat or any other object that can be assigned an Internet Protocol address and transfer data over a network.

AIoT enhances the capability of IoT by adding the power of machine learning algorithms to improve decision-making processes.

9.3.1 How Does AIoT Work?

In AIoT devices, AI is embedded into programs and infrastructure components, like chips, which are all connected using IoT networks. Efficient exchange of data amongst all hardware, software and platform components without human intervention is ensured using pre-written APIs.



Credit: monicaodo / Shutterstock

FIGURE 9.12 Autonomous Vehicle

IoT devices generate as well as collect data, that is then analysed using AI techniques to provide better insights into data. This improves efficiency and productivity. **Figure 9.12** shows an autonomous vehicle in which AI is embedded into chipsets, which are all connected using IoT networks.

AIoT data when processed using Edge AI (discussed later in the chapter) minimizes the bandwidth needed to move data while avoiding possible delays to data analysis. This further adds to efficiency of IoT.

9.3.2 Where Does AI Unlock IoT?

IoT is all about implanting sensors into machines. These sensors collect streams of data through Internet connectivity. All IoT-related services follow five basic steps to perform the assigned task—***Create, Communicate, Aggregate, Analyse and Act.*** Since Act depends on Analyse, the efficiency of any IoT system depends on the Analysis step. AI technology when used as the analysis stage, improves both efficiency and productivity manifold.

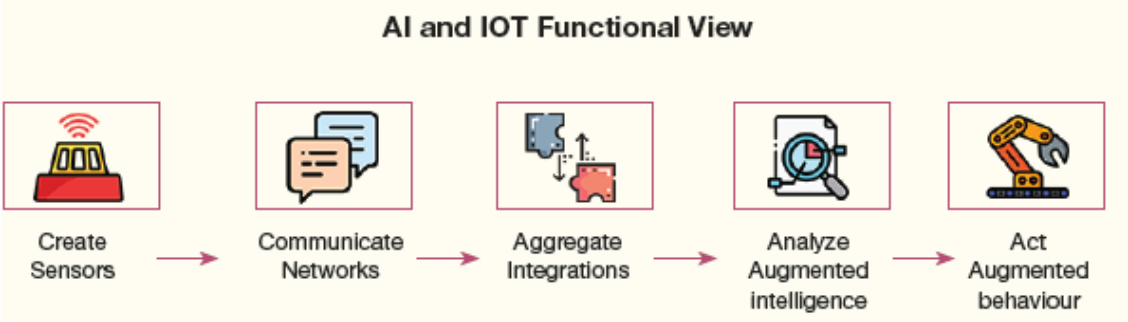


FIGURE 9.13 AI and IOT functional view

AIoT helps in achieving the following agile solutions:

1. Manage, analyse and obtain meaningful insights from data (refer **Fig. 9.13**)
2. Provide fast and accurate data analysis
3. Maintain a balance between requirements for localized and centralized intelligence
4. Protect data privacy and confidentiality
5. Ensure security against cyber attack

9.3.3 Applications and Examples of AIoT

AIoT is extensively used in the following domains.

1. **Smart cities:** Smart technology such as sensors, lights and meters collect data that is processed using AI to improve operational efficiency, drive economic growth and enhance quality of life for residents.
2. **Smart retail:** Smart cameras are now being used at retail outlets by managers to identify faces of customers who have scanned their items at the self-checkout counter before leaving the store.
   Moreover, cameras and sensors installed at multiple places to observe customers' movement and predict when they will reach the checkout line. This helps in maintaining dynamic staffing levels to enhance the productivity of the cashiers and reduce the customer's checkout time.
   Major retailers can use AIoT solutions to grow sales through customer insights. Data such as mobile-based user behaviour and proximity detection offer valuable insights to deliver personalized marketing campaigns to customers while they shop, increasing traffic in brick-and-mortar locations.

3. **Smart home:** In smart appliances installed at homes, data collected from human interaction with devices is analysed to understand user habits to provide customized support.
4. **Manufacturing:** These days, manufacturers use smart chips to detect when a particular equipment is not functioning properly, or when it is the right time to replace.
5. **Hiring:** AIoT technology when used to integrate data from social media and HR-related platforms can help to identify right talent without any bias.
6. **Autonomous vehicles:** These vehicles collect real-time data about nearby vehicles, pedestrians and other objects using multiple video cameras and sensors. Data is also analysed to monitor driving conditions and take appropriate decision instantly.
7. **Robots: Autonomous robots have multiple s**ensors installed to gather data about the environment. AI is then used to make decisions that help robot to make smart and optimized moves.
8. **Healthcare:** Medical devices and wearables collect and monitor real-time health data, such as heart rate. Moreover, healthcare facilities produce high volumes of data including patient information, imaging and test results. This information is valuable and necessary to take good care of the patient care. So, this data needs to accessed and processed quickly to suggest decision regarding patient's diagnostic and treatment.

   IoT combined with AI not only improves diagnostic accuracy, enable telemedicine and remote patient care but also reduces the administrative burden of tracking patient health in the facility.
9. **Smart Thermostat Solution:** The smartphone integration with smart thermostat solution can be used by users to check and manage the temperature from anywhere. Users can set the temperature based on their work schedule and temperature preferences. For example, Nest's smart thermostat solution uses AI-powered IoT.
10. **Drone Traffic Monitoring:** AIoT can be used by drones to monitor real-time traffic and make adjustments to the traffic flow to reduce congestion. While drones can collect and transmit traffic data, AI can be used to analyse that data and make decisions about how traffic can be regulated- make diversions, adjustments to speed limits and timing of traffic lights without human intervention.

For example, the ET City Brain, a product of Alibaba Cloud, uses AIoT to optimize the use of urban resources. It can detect accidents, illegal parking and can alter traffic lights to help ambulances reach patients/ hospitals faster.

11. **Office Buildings:** AIoT is used in smart office buildings. In such buildings, usually, a network of smart environmental sensors is installed that detect number of people present in the office and adjust temperatures and lighting accordingly to improve energy efficiency.
    These smart buildings are often accessed through facial recognition technology using a combination of connected cameras and AI to compare images taken in real-time against a database to determine who should be granted access. this technique is also used by employers to maintain attendance of employees for mandatory meetings.
12. **Fleet Management:** AIoT is used in fleet management to monitor a fleet's vehicles, reduce fuel costs, track vehicle maintenance and identify unsafe driver behaviour. Through IoT devices such as GPS and other sensors and an artificial intelligence system, companies are able to manage their fleet better thanks to AIoT.

9.3.4 Benefits and Challenges of AIoT

The benefits of AIoT include the following:

1. **With AI, operational efficiency of IoT is increased.** AIoT devices analyse data to reveal patterns and insights and automatically adjusts system operations.
2. **On the fly decision-making: With AIoT and edge computing, vital decisions can be taken instantly. Therefore, data generated by the devices are** analysed to identify points of failure/concerns to make appropriate adjustments as and when required.
3. **Reduced workload: Since data is analysed by AI and not humans, lot of time and money spent for this time is saved.**
4. **Scalability:** The number of devices connected to an IoT system can be increased anytime as per requirements to optimize existing processes or introduce new features. IoT devices range from mobile devices and high-end computers to low-end sensors. Low-end sensors offer most of the data. An AI-powered IoT ecosystem analyses and summarizes the data from one device before transferring it to other devices. This not

only reduces large volumes of data to a handy level but also enables connecting a large number of IoT devices.

5. **Boosting Operational Efficiency:** AIoT instantly analyses constant streams of data to predict the operational conditions and identify the parameters to be modified to obtain better results. Thus, intelligent IoT can deduce which processes are redundant and time-consuming, and which tasks can be fine-tuned to enhance efficiency. For example, Google uses AIoT to reduce its data centre cooling costs.
6. **Better Risks Management:** Pairing AI with IoT facilitates businesses to understand as well as predict a wide variety of risks and automate process to provide a prompt response. This helps them to handle financial loss, ensure employee safety and prevent potential cyber threats. For example, Fujitsu ensures safety of the employees by using AI for analysing data coming from connected wearable devices.
7. **Triggering New and Enhanced Products and Services:** With NLP, users can easily communicate with devices. With this human-machine interaction, IoT and AI can together be used to create new products and services. Even existing products and services can be enhanced by quickly processing and analyzing real-time data. For example, Rolls Royce uses AI technologies in the implementation of IoT-enabled airplane engine maintenance to identify patterns and gain useful operational insights.
8. **Eliminates Costly Unplanned Downtime:** In industrial manufacturing and in some fields like offshore oil and gas exploration, equipment breakdown can result in costly unplanned downtime. Predictive maintenance with AI enabled IoT predicts equipment failure well in advance and schedule orderly maintenance procedures. This reduces the side effects of downtime. For example, Deloitte, integrates AI and IoT for the following:
    1. Reducing 20%–50% of their time spent in maintenance planning
    2. Increasing 10%–20% of equipment availability and uptime
    3. Reducing 5%–10% amount spent as maintenance costs

All this portrays a rosy picture but do not forget that there are always two faces of the same coin. AIoT may fail. For example, autonomous delivery robots that fail might cause delays in product delivery; smart retail stores may lead to accidently stealing of a product if it fails to read a customer's

face, a road accident can be caused if an autonomous vehicle fail to read its surroundings, like a red light on traffic signal.

9.3.5 Future of AIoT

Integration of AI and IoT (refer **Fig. 9.14**) creates a much smarter system that makes accurate judgments without human intervention. AIoT will get a tremendous boost with 5G networks that is designed to enable faster transfer of large data files through higher bandwidth and lower latency. AIoT can revolutionize supply chains and delivery models and solve existing operational problems very efficiently. Several businesses have already adopted AI and IoT as part of their processes and products. In fact, AI and IoT are the top technologies in which companies are investing money to increase their operational efficiency and provide a competitive advantage. The graph in **Fig. 9.15** depicts the details.



Credit: elenabsl / Shutterstock



Credit: marigranula / 123RF

FIGURE 9.14 An AIoT Machine

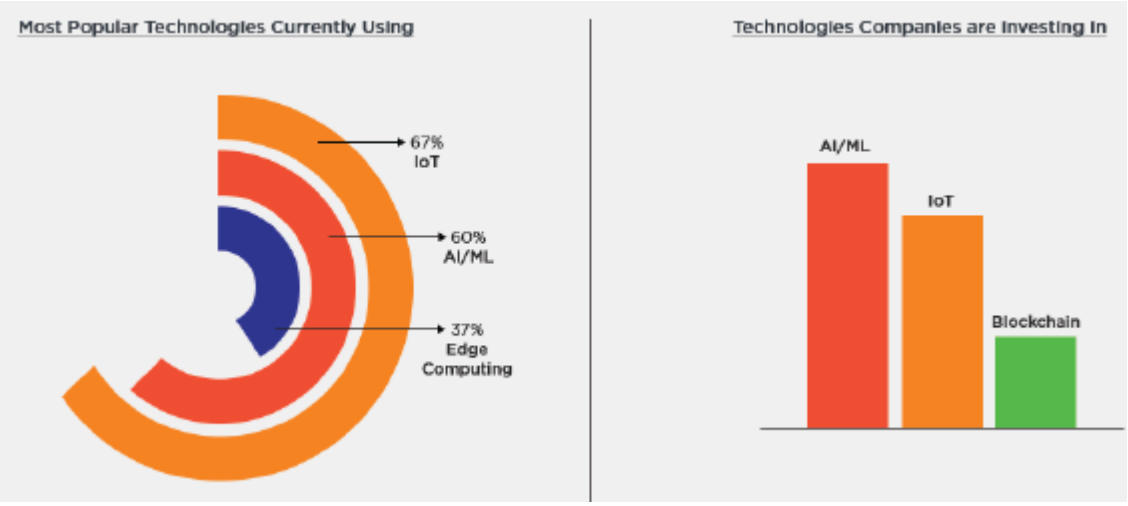


FIGURE 9.15 Visualization of emerging trends

It is surprising to note that every day, around one billion gigabytes of data is being generate by IoT devices. By 2025, it is estimated that 42 billion IoT-connected devices will be present globally and with growth in number of devices and network, the data generated is set to rise exponentially. With more data, there will be more challenges than opportunities.

With AI, IoT networks and devices can learn from past decisions, predict future activity, and continuously improve their decision-making capabilities and thus performance. AI allows the devices to 'think for themselves' by interpreting data without the delays and congestion that occur from data transfers.

9.4 Edge Computing

Edge computing is a distributed information technology (IT) architecture in which client data is processed at the periphery of the network, as close to the originating source as possible.

We have seen that data is the lifeline of any business as it provides valuable business insight and supports real-time control over critical business operations. Today, every business generates oceans of data, most of it coming from sensors and IoT devices operating in real time. This virtual flood of data is also changing the way computations are done. Traditional computing relies on centralized processing but bandwidth limitations, latency issues and unpredictable network disruptions poses a big challenge for hassle free movement of data and results. Edge computing offers a solution to overcome these challenges.

Edge computing moves some portion of storage and computations out of the central data centre and closer to the source of the data. This not only avoids transmission of raw data to central server but also facilitates data processing and analysis at the point where the data is actually generated —be it a retail store, a factory floor or a sprawling utility or across a smart city. Only the results of the computations that may include real-time business insights, equipment maintenance predictions or other actionable answers, is sent to the central processor for review.

Apart from efficiency benefits, edge computing also seems to be the only viable option available today. These days, the number of devices connected to the Internet, the volumes of data being produced by them is growing too fast to be handled by the traditional data centers. Gartner predicted that by 2025, 75% of the data will be created outside of centralized data centres. Movement of this big data amidst congestion and disruption is not possible.

So, IT architects have shifted focus from the central data centre to the logical *edge* of the infrastructure by taking storage space and computing resources from the data centre and moving them to locations where the data is generated. Therefore, they believe, ***'if you can't get the data closer to the data centre, get the data centre closer to the data.'***

A few applications that prefer edge computing include the following:

1. Retail environments combine video surveillance of the showroom floor with actual sales data to analyse the trend in consumer demand.
2. Predictive analytics that guide equipment maintenance and repair before actual defects or failures occur.
3. Utilities like water treatment or electricity generation.

Fog computing environments produce massive amounts of sensor or IoT data generated across large physical areas that are bigger than an edge. Examples include smart buildings, smart cities or even smart utility grids. In a smart city, data can be used to track, analyse and optimize the public transit system, municipal utilities, city services and guide long-term urban planning. A single edge deployment simply isn't enough to handle such a load, so here fog computing comes into picture.

9.4.1 Why is Edge Computing Important?

Consider the constant increase in demand of self-driving cars. These vehicles rely on intelligent traffic control signals which in turn need to produce, analyze and exchange data in real time. Multiply this requirement by huge numbers of autonomous vehicles to get a picture of how much data will be generated and processed in just one second. Clearly, this calls for a fast and responsive network and edge as well as fog computing addresses the concerns of a traditional computing system that include the following.

Bandwidth

Bandwidth refers to the amount of data that can be transmitted across a network over time. It is expressed in bits per second.

Every network has a limited bandwidth. Wireless networks have stringent limits of bandwidth as compared to wired networks. This means that only a limited data can be exchanged by a finite number of devices across the network.

However, network bandwidth can be increased to accommodate more devices and data but with a significant cost.

Latency

Latency is the time needed to transfer data from one point to another on a network. Ideally communication must take place at the speed of light, but it is often limited by large physical distances and issues like network congestion. Delays in data transfer adversely analysis and decision-making processes thereby hampering the ability for a system to respond in real time. In certain situations, latency can even cost lives. For example, in case of autonomous vehicles.

Congestion

The Internet is a global ‘network of networks’ that connects billions of computers all over the world to facilitate data exchange. Volumes of data transferred through these devices overwhelm the network with large number of data packets thereby increasing network traffic. Above this, network outages (or server/machine downtime) further add to these problems, making the Internet of Things useless.

By deploying servers and storage at a point near to the location where data is generated, edge computing (refer **Fig. 9.16**) can enable many devices to communicate over a much smaller and more efficient LAN that provides sufficient bandwidth making latency and congestion issues virtually nonexistent.

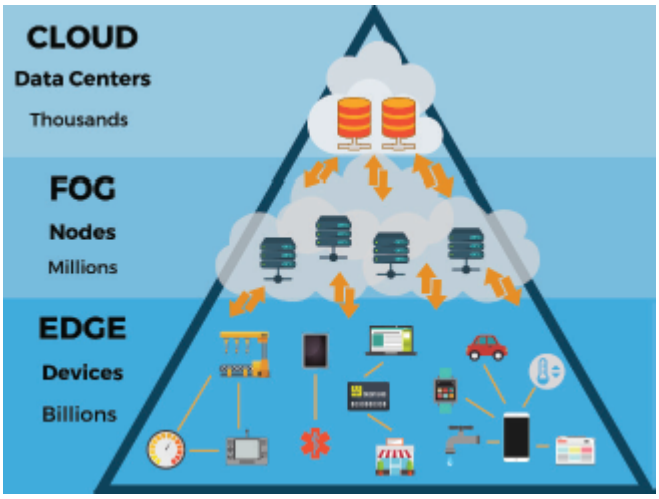


FIGURE 9.16 Edge Computing

Edge computing can reduce network costs, avoid bandwidth constraints, reduce transmission delays, limit service failures and provide better control over the movement of sensitive data.

In edge computing, local storage collects and protects the raw data, local servers perform essential data processing and analysis to make decisions in real time and the send the results, or just essential data, to the cloud or central data centre.

9.4.2 Edge Computing Use Cases and Examples

Technically, edge computing collects, filters, processes and analyses data 'in-place' at or near the network edge. Being a powerful technique to address vital issues including network bandwidth and congestion, it is being used in several domains including the following:

Manufacturing

Edge computing is used by manufacturers to monitor the manufacturing process, enable real-time analytics and identify production errors to improve quality of the products. Edge computing also supports addition of environmental sensors throughout the manufacturing plant to provide insight into how each product component is assembled and stored.

Manufacturing units also use this technology to make faster and more accurate business decisions regarding its operations.

Farming

Edge computing is used for indoor farming to make efficient use of sensors to track water use, nutrient density and determine optimal harvest. Data is collected and analysed to find the effects of environmental factors and improve the crop growing algorithms and ensure that crops are harvested in peak condition. Without edge computing it is very challenging to grow crops indoors without sunlight, soil or pesticides. The process reduces grow times by more than 60%. Hence, edge computing is a ray of hope to produce better results.

Network Optimization

Edge computing assists in optimizing the performance of a network by analysing data and identifying the most reliable, low-latency network path for each user's traffic.

Workplace Safety

Edge computing can combine and analyse data from on-site cameras, employee safety devices and various other sensors to help businesses keep a check on workplace conditions or ensure that employees follow established safety protocols. This is more important when either the workplace is at a remote location or is unusually dangerous (like construction sites or oil rigs).

Improved Healthcare

The healthcare industry generates massive amount of patient data that is often collected from devices, sensors and other medical equipment. This huge chunk of data requires edge computing to automate data storage and processing. Machine learning algorithms are used to access the data and extract information that requires immediate action to help patients avoid health incidents in real time.

Transportation

Autonomous vehicles require and produce anywhere from 5 TB to 20 TB every day. Data about location, speed, vehicle condition, road conditions, traffic conditions and other vehicles needs to be stored and analysed in real time, while the vehicle is in motion. In such a scenario, autonomous vehicle becomes an ‘edge’ to facilitate authorities and businesses manage vehicle fleets based on actual conditions on the ground.

Retail

Retail businesses generate enormous data volumes from surveillance, stock tracking, sales data and other real-time business details. In such a situation, edge computing helps to analyse this diverse data and identify business opportunities like an effective endcap or campaign, predict sales and optimize vendor ordering. Since retail businesses vary from one location to another, edge computing is preferred for processing the data locally at each store.

Immersive Experience

Real-time applications, such as AR/VR, cannot tolerate more than a few milliseconds of latency and can be extremely sensitive to jitter, or latency variation. In many applications, closed-loop automation is used to maintain high availability with response times in tens of milliseconds. This stringent requirement cannot be fulfilled without edge computing infrastructure.

Edge computing expands bandwidth capabilities thereby supporting new immersive applications including AR/VR, 4K video, and 360° imaging for verticals like healthcare. Caching and optimizing content at the edge is already becoming a necessity since protocols like TCP does not perform well when there are sudden changes in radio network traffic.

Edge computing infrastructure when used to access real-time data through network can reduce stalls and delays in video by up to 20% during peak viewing hours.

Education

Edge computing facilitates software-based education solutions that use on-device AI for personalized virtual assistance, natural language interaction and even augmented reality experiences. For example, ViewSonic digital whiteboard experience uses edge technology to re-create the classroom experience for students and teachers engaged in distance learning.

9.4.3 Benefits of Edge Computing

Besides solving critical issues like bandwidth limitations, excess latency and network congestion, edge computing also offers following benefits to the users.

Autonomy

In cases where connectivity is unreliable or bandwidth is restricted due to location's geography, edge computing is a viable solution. For example, oil rigs, ships at sea, remote farms, remote locations like a rainforest or desert extensively used edge computing to compute work on site, sometimes on the edge device itself. Like in a water purifier used in a remote village, that has water quality sensors installed in it, edge computing can save data in its local storage and then transmit it to a central point only when connectivity is available. By processing data locally, the amount of data to be sent is reduced significantly, thereby demanding less bandwidth or connectivity time than was required otherwise.

Data Sovereignty

Transmitting large volumes of data not only adds to technical woes but also poses a great threat to its security, privacy and other legal issues, especially when data crosses national and international boundaries. Edge computing keeps data close to its source and within the bounds of prevailing data sovereignty laws which defines how data should be stored, processed and exposed. Processing raw data locally secures sensitive data before it is exchanged with the cloud or primary data center (which may be in other jurisdictions).

Edge Security

Edge computing ensures data security. Since data and results obtained from analysing the data has to travel to reach the cloud or data center, steps are taken to protect the data through encryption. Edge deployment is often hardened against hackers and other malicious activities.

Moreover, edge computing can be thought of as similar to data centre computing due to the following reasons:

1. It includes compute, storage and networking resources.
2. Its resources may be shared amongst many users across several applications.
3. It can be benefitted by virtualization and abstraction of the resource pool.
4. It supports the ability to leverage commodity hardware.
5. It uses APIs to support interoperability.

9.4.4 Challenges of Edge Computing

Although edge computing provides a range of benefits across a multitude of use cases, the technology is far from foolproof. There are certain issues that can affect the adoption of edge computing at a large scale. These include the following.

Limited Capability

Edge computing deploys a variety of resources to provide a range of services. But this deployment of infrastructure at the edge must be clearly defined and justified to be effective. This is because even an extensive edge computing deployment serves a specific purpose at a pre-determined scale.

Connectivity

Edge computing still needs to transfer data to the cloud and hence even its capability is somewhat restricted by network connectivity.

Security

IoT devices are notoriously insecure, so edge computing deployment needs to take extra care for proper device management, policy-driven configuration enforcement and enforcement of security techniques in the computing and storage resources. Software patches and updates along with encryption ensure secure communications.

Data Lifecycles

Today, a lot of data is generated and collected by IoT devices. But most of this data is not required after a short span of time. So, it may not be stored for a long term. Therefore, it is necessary that the business has the capability to decide which data to keep and what to discard once the analyses process is over. The business also has the responsibility to retain and protect the required data in accordance with business and regulatory policies.

Greater Demands from Local Hardware

Edge computing requires more powerful and robust local hardware. For example, in a simple IoT camera a built-in computer is used to send its raw video data to a web server. But with edge computing, a more powerful computer with more processing power is required to run its own motion-detection algorithms.

Difficult to Scale

Scaling out edge servers to many small sites is more complicated than adding the equivalent capacity to a single data centre. Moreover, increase in physical locations also adds overhead to manage them.

Limited Support

Edge computing sites are usually remote with limited or non-availability of on-site technical expertise. If something fails on site, there must be an infrastructure in place that can be fixed easily by non-technical local labour. Further issues can be addressed centrally by a small number of experts located elsewhere.

Issues in Site Management

It takes more time, money, resources and efforts to manage all sites that are a part of edge computing. It is more challenging when software is implemented differently at each site.

More Prone to Attacks

Physical security of edge sites is often much lower than that of the central site. So, there is a greater risk of malicious or accidental attacks.

9.4.5 Edge Computing Implementation

Edge computing seems to be a straightforward idea but implementing it is a challenging exercise.

The first important step of deploying any technology successfully is the creation of a meaningful business and technical edge strategy which considers the need for edge computing. The ‘why’ answer demands a clear understanding of the technical and business problems that the organization needs to solve—overcome network constraints, maintain data sovereignty and so on.

An edge data center requires careful upfront planning and migration strategies.

The edge strategy starts with what the edge means, where it exists for the business and how it should benefit the organization. It should align with existing business plans and technology roadmaps.

As the project moves closer to implementation, evaluate hardware and software options carefully. Select a vendor by going through each vendor’s offering and evaluate it based on cost, performance, features, interoperability and support. The software tool should provide comprehensive visibility and control over the remote edge environment. Some of the vendors in edge computing are Adlink Technology, Cisco, Amazon, Dell EMC and HPE.

The actual deployment of an edge computing varies dramatically in scope and scale. No two edge deployments are the same. Because of these variations, edge strategy and planning become critical to the success of an edge project.

An edge deployment demands comprehensive monitoring. Deploying IT staff for this may not be possible, so edge deployments should be architected to provide tools that satisfy the following:

1. Ensure resilience, fault-tolerance and self-healing capabilities
2. Can be deployed remotely
3. Easy to configure
4. Offers comprehensive alerting and reporting about site availability or uptime, network performance, storage capacity and utilization and compute resources
5. Maintains security of the installation and its data. Tools that emphasize vulnerability management and intrusion detection and prevention should be installed.

9.4.6 Edge Computing, IoT and 5G Possibilities

Edge computing is evolving with new technologies and edge services are expected to become available worldwide by 2028. Today, edge computing is situation-specific, but it is expected to become more ubiquitous and support much more use cases in the future.

More number of devices and software (compute, storage and network) will be developed for edge computing.

Multivendor partnerships will provide better product interoperability and flexibility at the edge. For example, AWS and Verizon will together bring better connectivity to the edge.

Wireless communication technologies, such as 5G and Wi-Fi 6, will also boost edge deployments and utilization in the coming years. This will set the way for better and enhanced virtualization and automation capabilities, better vehicle autonomy and workload migrations to the edge, while making wireless networks more flexible and cost-effective.

The future will focus on the development of micro modular data centres (MMDCs). The MMDC is basically a data centre in a box, putting a complete data centre within a small mobile system that can be deployed closer to data—across a city or a region, to bring computing much closer to data.

Moreover, while AI algorithms that require large amounts of processing power are run on cloud-based services, AI chipsets that can do the work at the edge will help computations to be handled by edge systems.

Network function virtualization (NFV) is at the heart of edge computing application as it provides infrastructure functionality. Telecom operators are now transforming their service delivery models by running virtual network functions as part of, or layered on top of, an edge computing infrastructure.

9.5 Metaverse

The term ‘metaverse’ is a combination of two words—‘meta’ and ‘universe’ which means an anticipated future iteration of the Internet that is often referred to as Web 3.0. The evolution of the Internet from Web 2.0 to Web 3.0 will cause a tremendous boost of online 3-D or virtually integrated environments that provide users immersive experience in true sense (refer **Fig. 9.17**).



Credit: marigranula / 123RF

FIGURE 9.17 Metaverse

Metaverse is a combination of the virtual reality, augmented reality, mixed reality, blockchain technology, Internet of Things (IoT), 5G, **artificial intelligence (AI)** and spatial technologies that is accessed through a browser or headset. It is a device-independent technology that is not owned by a single vendor. Metaverse is an independent virtual economy that uses digital currencies and **non-fungible tokens (NFTs).** It allows users to have real time interactions and experiences across distance.

Basically, metaverse is not just one type of experience. It actually encompasses a continuum of immersive digital experiences that will be available to users in the future to engage them in different activities in digital spaces. For example, a massive virtual reality multiplayer game can be accessed through a VR headset. Similarly, an integrated digital and physical spaces (like location-specific immersive digital content) that are often a need of business users, can be experienced using digital glasses or smartphones.

Many social media and tech companies like Meta Platforms (formerly known as Facebook) and Microsoft are investing heavily in Social VR to create platforms to allow people to connect socially and work remotely, respectively.

Metaverse is not one digital space but a combination of several digital spaces and experiences.

9.5.1 Use Cases of Metaverse

Metaverse is considered to be the next version of the Internet. The Internet started as individual bulletin board are rose to independent online destinations. Metaverse will make these destinations sites on a virtual shared space. Some examples are listed below.

Nikeland is a place to hang out, play and dress your avatar in virtual Nike products. More than 7 million people have already visited Nikeland.

In 2020, 12.3 million people attended a virtual concert by rapper Travis Scott, hosted in Fortnite. In this concert, the dancing figures were all people, connecting from different locations around the world.

The best part of Metaverse is that it allows users to mimic being together in ways that are not possible over Zoom. You can point to something to explain, use hand gestures or draw on a piece of paper or visit places together.

Metaverse makes it possible to collaborate with surgeons, or create a clay model for a new-car design.

Users can even purchase outfits and accessories for online avatars or buy digital land and construct virtual homes.

Virtual meetings in metaverse gives you the feeling of being together in a room. When participants need a work break, they can together play virtual games with colleagues from all over the world. This creates a real connection and builds social bonds.

Social media can move to the Metaverse, where users can interact through three-dimensional avatars.

Higher education, medical, military and other types of training can be delivered in a more immersive learning experience without creating additional infrastructure. Metaverse provides the framework to provide these services.

‘Metaverse will allow people to replicate or enhance their physical activities. This could happen by transporting or extending physical activities to a virtual world or by transforming the physical one,’ says Marty Resnick

Virtual malls use metaverse to extend their reach to an immersive shopping experience.

As of now, some aspects of the metaverse are integrated into existing Internet-enabled video games (like Second Life, Minecraft and Fortnite). These games offer immersive experiences in which players from around the world can engage simultaneously.

Did you know?

1. Microsoft is working hard to integrate ***Microsoft Teams into the metaverse.***
2. Roblox, an online gaming platform (particularly for youngsters), is creating an integrated set of virtual experiences for gamers.
3. Nvidia Corp is building an Omniverse platform that connects 3D virtual words in a shared universe. This can be used for creating simulations of buildings and factories.
4. In 2021, Snapchat launched AR glasses to augment users' experience of reality outside their app.
5. Epic Games developed one of the popular digital game- Fortnight. It is now creating virtual social experiences like music concerts or dance parties.

Gartner estimated 25% of people will spend at least one hour every day in the Metaverse for work, shopping, education, social media and/or entertainment by 2026.

Woolsey said, an organization must go for metaverse solution if the following conditions are satisfied.

1. The target customers are 15–30-year-olds. Metaverse engages the audience and makes them stay consistent with the brand. Businesses can even build a store that sells virtual goods or deliver an event or other type of engagement.
2. The organization has a dispersed team and often goes for hybrid meetings, multi-location training programmes and institutional events that bring people together to build a community, affiliation and engagement.
3. The organization needs to explore its internal operations—industrial or operational scenarios that would otherwise be very expensive to build in real life. **Figure 9.18** shows some key elements of metaverse.

Auto companies usually use digital twins to perform a dummy crash test in a metaverse world, and then collaborate on model modifications in augmented reality.

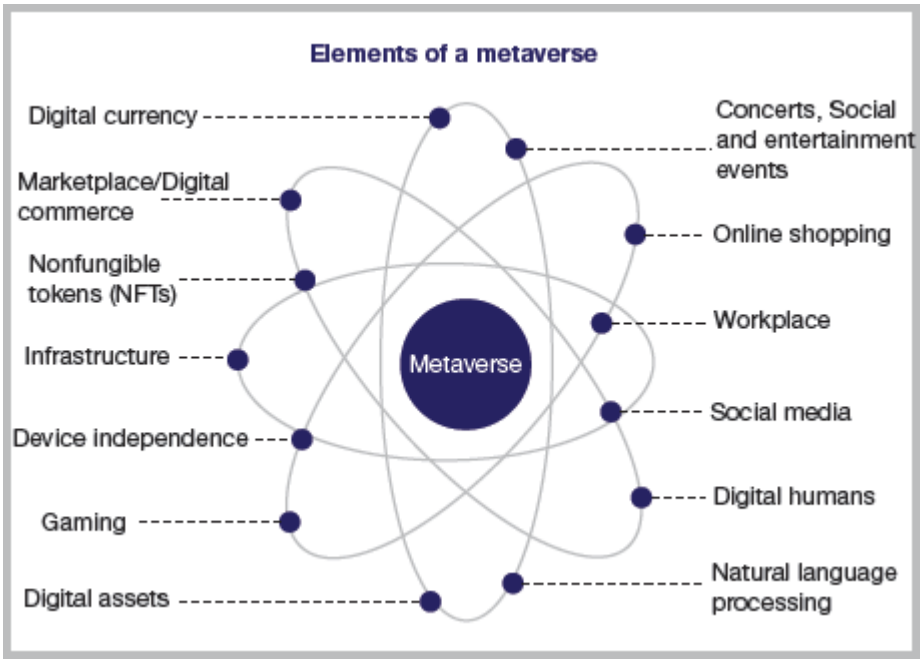


FIGURE 9.18 Elements of a metaverse

All in all, metaverse is seen as a significant business and financial opportunity for the tech industry and other sectors.

Woosely suggested that an organization must not dive into metaverse, just for the sake of doing metaverse. It should first analyse if any of the above-stated three scenarios are benefitted with this technology.

Once the technology is ready, first experiment it with a small audience so you can understand its actual value both qualitatively and quantitatively. Based on this learning, find the path that delivers strategic value and can be easily scaled.

However, the adoption of Metaverse technologies is still nascent and fragmented. Not many companies are planning to invest heavily in it, since it is still too early to determine which investments will be viable in the long term.

AR is used in Google Search to allow users to place 3D objects in their own space, allowing them to gain a better idea of an object's scale.

Many critics are skeptical about the growth of metaverse as it has mostly failed to live up to its hype and revenue expectations. Moreover, critics have exhibited concerns about the privacy and health implications of the metaverse vision. So, instead of investing, companies are first learning, exploring and preparing for a Metaverse.

Recommendations for Technology Innovation Leaders

1. Develop digital business strategies that leverage the built-in infrastructure and participants of the Metaverse.
2. Lead ideas and innovation that focuses on new opportunities and business models with the Metaverse.
3. Identify risks, privacy and security issues that are prevalent in this new persistent and decentralized environment.
4. Enumerate the outcomes, opportunities and obstacles faced by Metaverse.

Side-Effects of Metaverse

According to Indo-Asian News Service (IANS) report, Meta (formerly known as Facebook) is aware that VR can be a really 'toxic environment', especially for women and minorities. Harassment in digital spaces is not uncommon.

India's Role in Building Metaverse

The IANS report revealed that Mark Zuckerberg said that India and its vast pool of talent—the engineers, developers and creators and the whole vibrant startup ecosystem, is going to be a huge part of that journey. He is really excited about the role that India will play in building the future of Metaverse, which is going to be the successor to the mobile internet. India will have the largest app developer base in the world by 2024. It already has one of the largest 'Spark AR' developer communities.

9.5.2 Where is Metaverse Today?

As of now, Meta owns the leading producer of VR headsets, Oculus. Microsoft supports various enterprise use cases with its HoloLens MR smart glasses that ere first released in 2016. Apple will be soon releasing its AR and VR headsets. HTC, Pico, MagicLeap and other manufacturers are also moving to manufacture devices to support this technology. At the same time, cascade technology architectures are shifting various computational loads across back-end server infrastructures to edge devices. For example, Unity's streams fully interactive real-time 3D environments. The job of rendering the environments is handled by their automatically scaling GPU server infrastructure.

Metaverse makes full use of ubiquitous computing experience to allow users leverage traditional devices like computers and mobile devices while experiencing with emerging immersive AR and VR wearables.

Like the internet, no one owns the metaverse. However, there are several key players in the space, and companies like Meta, Microsoft, Unity, Epic Games, Roblox and others are pouring billions of dollars into making the sci-fi dream come true.

The idea of metaverse was first coined by sci-fi author Neal Stephenson in his novel *Snow Crash*. Soon, it could become a reality because many large technology companies are betting that it will soon be as big as the Internet is today.



KEY TERMS

Internet of Things (IoT): A system of interrelated computing devices, machines, objects, animals or people that have unique identifiers for identification and also the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

Sensor: A device that detects and responds to some type of input that it receives from the physical environment.

Metaverse: A combination of the virtual reality, augmented reality, mixed reality, blockchain technology, Internet of Things (IoT), 5G, artificial intelligence (AI) and spatial technologies that is accessed through a browser or headset.

Edge computing: A distributed information technology (IT) architecture in which client data is processed at the periphery of the network, as close to the originating source as possible.

Expert system: A computer program that solves complex problems in a particular domain, at the level of extra-ordinary human intelligence and expertise.



RECAPITULATION POINTS

1. The user interface of an expert system must be very simple and intuitive so that even a non-expert user can communicate with the expert system to find solution to a complex problem.
2. Forward chaining starts from the known facts and rules. Backward chaining strategy is used for discovering cause or reason. For example, diagnosis of blood cancer in humans
3. AIoT enhances the capability of IoT by adding the power of machine learning algorithms to improve decision-making processes. IoT devices generate as well as collect data, that is then analysed using AI techniques to provide better insights into data.
4. The huge availability of address space in IPv6 has led to the development of the Internet of Things
5. Edge computing can reduce network costs, avoid bandwidth constraints, reduce transmission delays, limit service failures and provide better control over the movement of sensitive data.



REVIEW QUESTIONS

1. List the challenges faced by IoT devices.

2. What are the different types of sensors?
3. Why is Metaverse considered to be the next version of the Internet?
4. Write a short note on AIOT.
5. Discuss the capabilities, merits and demerits of an expert system.
6. List the steps involved in developing an expert system.
7. What challenges are faced by IoT today?
8. Explain the relevance of edge computing.
9. What is an expert system? Give its characteristics.
10. Explain the different components of an expert engine.
11. Explain the capabilities of an expert system.
12. Give any five applications of an expert system.
13. List the steps involved in developing an expert system.
14. Give some applications where AIoT and edge computing can be used.
15. Differentiate between the following:
    1. Forward and backward chaining
    2. Conventional system and expert system
    3. Human expert and artificial expert



FILL UP

1. A _____ is a device that detects and responds to some type of input that it receives from the physical environment.
2. All IoT-related services follow five basic steps to perform the assigned task—_____, _____, _____, _____ and _____.
3. Auto companies usually use _____ to perform a dummy crash test in a metaverse world.
4. A user interacts with expert system through the _____.
5. _____ chaining starts from the known facts and rules.
6. _____ is used in organic chemistry to detect unknown organic molecules.
7. _____ contains information and rules of a particular domain or subject.
8. _____ **knowledge** is based on practice, probability, evaluation and experiences.
9. _____ can lock out users from various devices.

10. _____ sensors detect forces per unit area in gases or liquids.
11. _____ sensors detect noise and measure distance between two objects.
12. _____ is the time needed to transfer data from one point to another on a network.



STATE TRUE OR FALSE

1. Expert systems are generic software.
2. More the amount of knowledge stored in the KB, better is the performance of the expert system.
3. CaDeT expert system could select specific software to generate a customized computer system based on user's requirements.
4. An expert system can deduce new facts from existing facts.
5. Metaverse is owned by Facebook.
6. More the amount of knowledge stored in the KB, better is the performance of the expert system.
7. Inference engines can also include explanation and debugging abilities.
8. Shell is an expert system without knowledge base.
9. Probabilistic Inference engine uses facts and rules to draw conclusions.
10. Forward chaining strategy is used for discovering cause or reason.
11. Results obtained by an expert system are highly reproducible.
12. The huge availability of address space in IPv6 has led to the development of the Internet of Things.
13. Particle sensors sense the movement or stoppage of parts, people, etc.
14. Metaverse allows users to interact through three-dimensional avatars.
15. Physical security of edge sites is often much lower than that of the central site.
16. Edge computing requires less powerful local hardware.



MULTIPLE CHOICE QUESTIONS

1. _____ is often referred to as Web 3.0.
    1. AIoT
    2. Edge Computing
    3. Metaverse
    4. IoT
2. _____ computing is a distributed IT architecture in which client data is processed at the periphery of the network, as close to the originating source as possible.
    1. Cognitive
    2. Affective
    3. Edge
    4. Transfer
3. Which expert systems was designed to find the bacteria causing infections like bacteraemia and meningitis?
    1. DENDRAL
    2. MYCIN
    3. PXDES
    4. CaDeT
4. _____ is an expert system that determines the type and level of lung cancer by analyzing picture(s) from the upper body.
    1. DENDRAL
    2. MYCIN
    3. PXDES
    4. CaDeT
5. _____ knowledge is based on practice, probability, evaluation and experiences.
    1. Factual
    2. Heuristic
    3. Both of these
    4. None of these
6. Identify the incorrect statement about expert systems.
    1. Expert systems cannot produce accurate output for inadequate knowledge base

2. Expert systems cannot refine knowledge base
3. Expert systems can substitute human decision makers
4. An expert system can predict results using past data.

7. Identify which of the following is not a characteristic of an expert system.
    1. Reliable
    2. Highly responsive
    3. Limited memory
    4. Reduced cost
8. _____ is the brain of the expert system.
    1. User interface
    2. Inference engine
    3. Knowledge base
    4. CPU
9. _____ represents that knowledge in the format accepted by the knowledge base.
    1. User
    2. Domain expert
    3. Knowledge engineer
    4. Any of these
10. What can an expert system do?
    1. Possess human capabilities
    2. Produce inaccurate output for inadequate knowledge base
    3. Refine knowledge base
    4. Use emotions to make decisions
11. _____ sensors sense objects passing within their field of detection.
    1. Photoelectric
    2. Radiation
    3. Proximity
    4. Position



ANSWERS

Fill Up

1. Sensor
2. Create, Communicate, Aggregate, Analyse and Act
3. digital twins
4. user interface
5. Forward
6. DENDRAL
7. knowledge base
8. Heuristic
9. Ransomware
10. Pressure
11. Ultrasonic
12. Latency

State True or False

1. False
2. True
3. False
4. True
5. False
6. True
7. True
8. True
9. False
10. False
11. True
12. True
13. False
14. True
15. True
16. False

Multiple Choice Questions

1. c
2. c
3. b
4. c
5. b
6. c
7. c
8. b
9. c
10. b
11. a